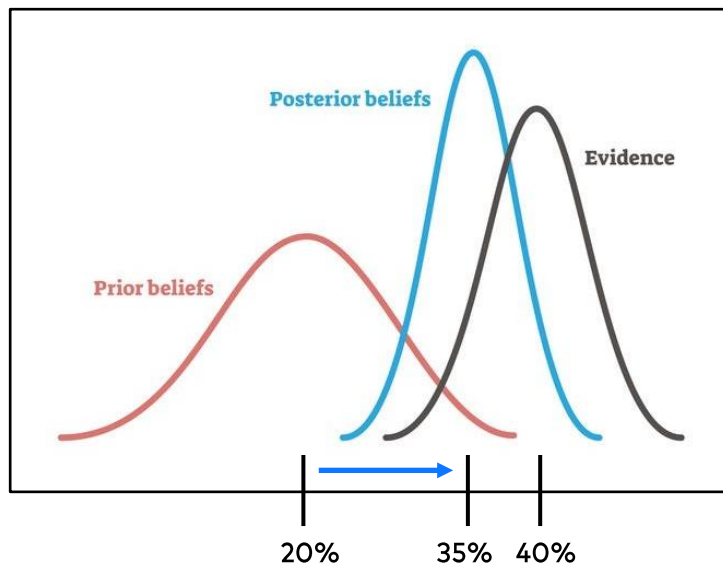


MODEL METHODOLOGY

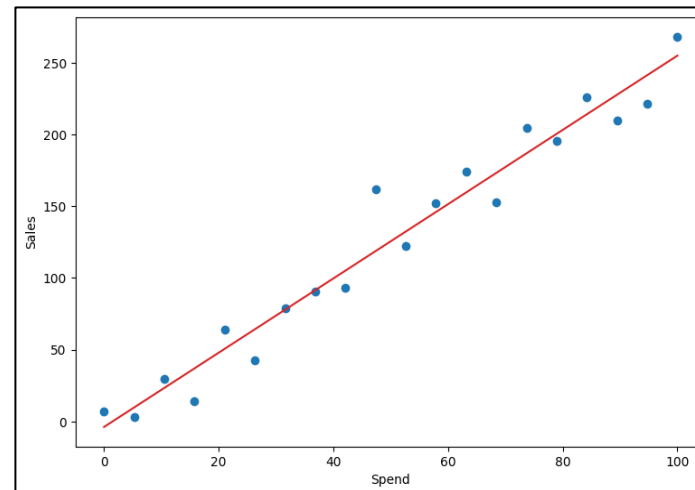
How does MMM work?

- It uses past data (priors) to make guesses about what's likely happening now and adjusts estimates based on evidence
- It looks for patterns in the data (kind of like drawing the best-fit line through a scatter plot) to figure out what's really driving results.
- It tells us how much impact each marketing channel's spend has on average.

Believed contributions of spend on attendance



Spend vs sales with line of best-fit



Equation to determine sales

$$Y_i = \beta_0 + \beta_1 X$$

Diagram illustrating the equation to determine sales:

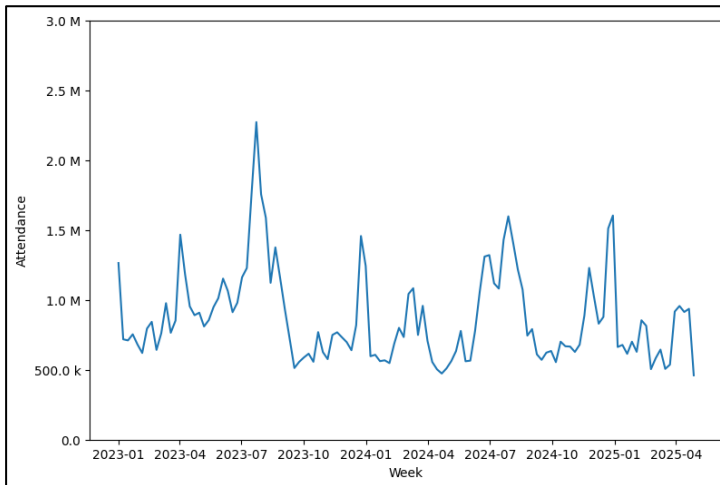
- Y_i is labeled Sales (with an upward arrow).
- β_0 is labeled Do-Nothing Sales (with a downward arrow).
- β_1 is labeled Sales multiplier (with an upward arrow).
- X is labeled Spend (with a downward arrow).

MODEL METHODOLOGY

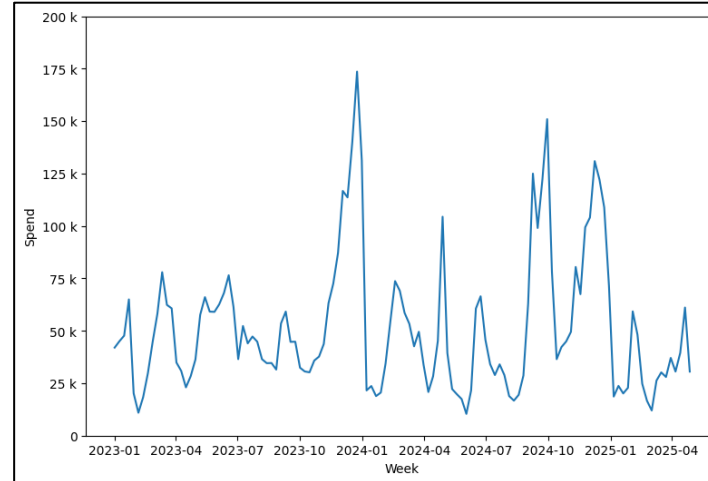
What data does MMM require?

- **KPI:** Attendance, Box Office Revenue, LBE Revenue, Revenue per Attendee
- **Paid channels:** SEM, Meta, Social Media, Programmatic
- **Organic channels:** Email and SEO
- **Control variables:** Weather, Holidays, Special Promotion

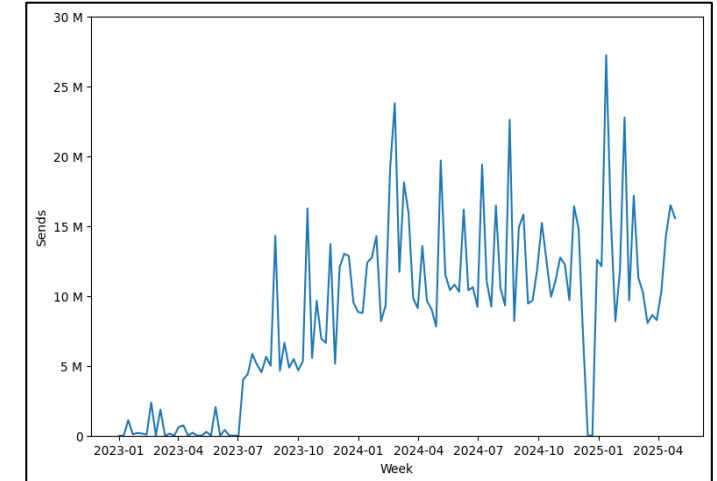
KPI (attendance)



Paid channel spend (total)



Organic channel sends (email)

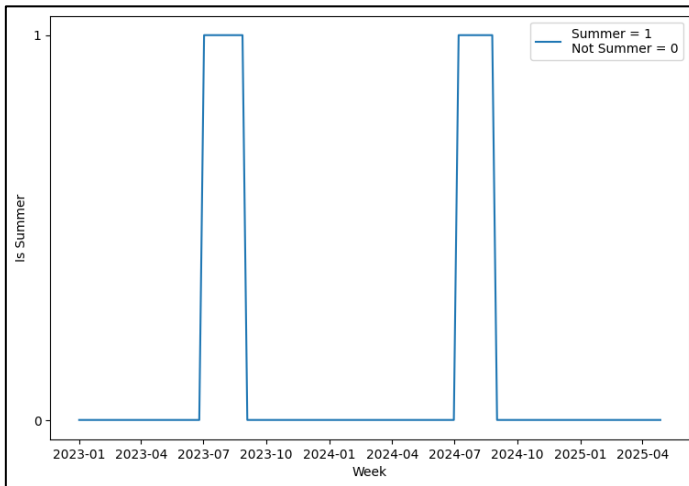


MODEL METHODOLOGY

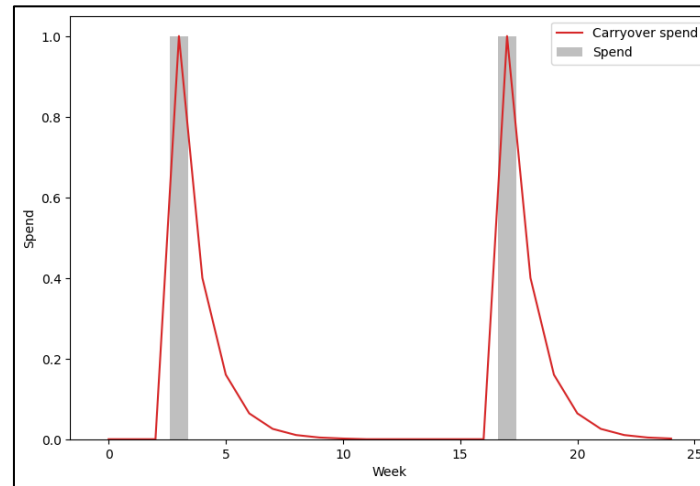
What assumptions are made?

- **Baseline Sales:** Even without marketing, there is a steady level of sales. This is assumed to be positive in the model.
- **Control Factors:** Other influences on sales (like summer season or holidays) help boost results, but they don't completely take over.
- **Marketing Memory (Adstock):** About 40% of marketing impact carries over to the next week, meaning campaigns have a moderate effect.
- **Diminishing Returns (Saturation):** Diminishing returns are seen early. As there is more spend, each extra dollar has less impact.

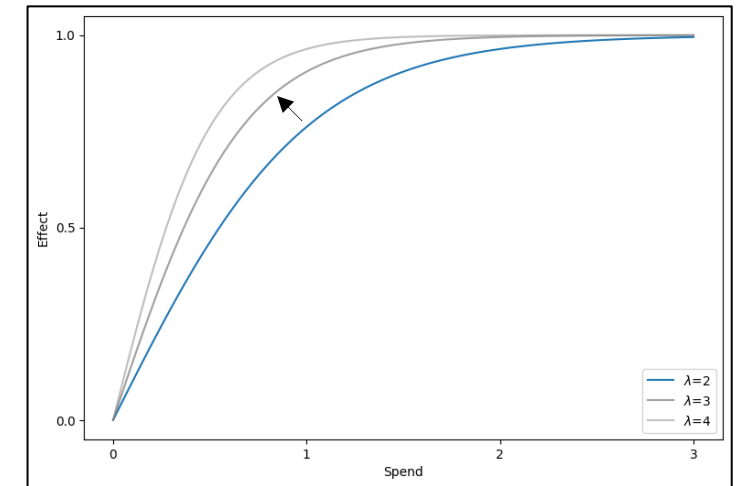
Seasonality variable



Spend and carryover by week



Spend saturation function



MODEL METHODOLOGY

How are the model results interpreted and calculated?

- Comparing contributions helps identify high-performing channels and underperformers. Higher contribution proportions mean higher efficiency of a channel relative how much was spent.
- The model finds the incremental lift each channel provides by simulating thousands of random values (e.g., picking out marbles from a bag with an unknown mix of colors, but we can assume what the mix could be based on past knowledge)
- This accounts for uncertainty and provides the most likely range of values.

$$Y_i = \beta_0 + \beta_1 X$$

Spend
↓
↑
Sales multiplier

